Summary

In this astrophysics topic learners will look in more detail at how we can investigate the characteristics of planets. To begin with learners will investigate bodies that are close to our own planet and consider factors that affect natural and artificial satellites. The topic then moves onto considering bodies within the universe, and will apply their knowledge of fusion processes to understand the life cycle of a star and waves to consider black body radiation. The Big Bang theory will be studied and the evidence that supports it as a scientific theory.

Underlying knowledge and understanding

Learners should already be familiar with the bodies within our own solar system and the behaviour of satellites. They may have a basic understanding of the Big Bang theory and that distances to other celestial bodies are large.

Common misconceptions

A common misconception among learners is that the Sun is not a star but a separate entity; it needs to be instilled in learners that the sun is a star. In addition, learners have difficulty grasping how far away celestial objects are.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Topic content			Opportunities to cover:		Practical suggestions
Learning outcomes		To include	Maths	Working scientifically	
P8.3a ☑	explain the red-shift of light as seen from galaxies which are receding (qualitative only). The change with distance of each galaxy's speed is evidence of an expanding universe	understanding of changes in frequency and wavelength		WS1.1b	Use of a Doppler ball to model red-shift. Use of a balloon to illustrate why galaxies are moving away from us and that expansion is from the centre of the universe.
P8.3b ☑	explain how red shift and other evidence can be linked to the Big-Bang model	CMBR			

P8.3c ☑	recall that our Sun was formed from dust and gas drawn together by gravity and explain how this caused fusion reactions, leading to equilibrium between gravitational collapse and expansion due to the energy released during fusion	lifecycle of a star		WS1.1a, WS1.1b, WS1.1c	Research and produce a poster illustrating the life cycle of a star.
P8.3d ☑	explain that all bodies emit radiation, and that the intensity and wavelength distribution of any emission depends on their temperatures	an understanding that hot objects can emit a continuous range of electromagnetic radiation at different energy values and therefore frequencies and wavelengths		WS1.1a, WS1.1b, WS1.1c, WS1.1d, WS1.1f, WS1.1g, WS1.1i, WS1.3e	Comparison of temperature changes inside sealed transparent containers with different gases inside. Research evidence of global warming from the last 200 years.
P8.3e ☑	recall the main features of our solar system, including the similarities and distinctions between the planets, their moons, and artificial satellites	the 8 planets and knowledge of minor planets, geostationary and polar orbits for artificial satellites and how these may be similar to or differ from natural satellites		WS1.1a, WS1.1b, WS1.1c, WS1.1g, WS1.1i	Building a model of the solar system to demonstrate scale. Research the evidence for the presence of the Moon as a result of a collision between the Earth and another planet. Research the uses of geostationary and polar satellites.
P8.3f ☑	**explain for circular orbits, how the force of gravity can lead to changing velocity of a planet but unchanged speed (qualitative only)**				
P8.3g ☑	**explain how, for a stable orbit, the radius must change if this speed changes (qualitative only)**				

P8.3h ☑	**explain how the temperature of a body is related to the balance between incoming radiation absorbed and radiation emitted; illustrate this balance using everyday examples and the example of the factors which determine the temperature of the Earth**	**an understanding that Earth's atmosphere affects the electromagnetic radiation from the Sun that passes through it**			
P8.3i ☑	**explain, in qualitative terms, how the differences in velocity, absorption and reflection between different types of waves in solids and liquids can be used both for detection and for exploration of structures which are hidden from direct observation, notably in the Earth's core and in deep water**	**P and S waves, use of sonar**	M5b	WS1.1a, WS1.1b, WS1.1c, WS1.1f, WS1.1h, WS1.3b	Examination of seismographic traces of recent earthquakes. Research the design of buildings that are in countries that experience earthquakes regularly and how the design is linked to P and S wave characteristics.